

SATS User Guide

Version: SATS 1.0.8

SATS (Signature Analyzer for Targeted Sequencing) is an R package for mutational signature analysis in targeted sequencing data. The method models panel-specific mutation opportunity, so it can be used for de novo signature detection, mapping of de novo profiles to tumor mutational burden (TMB)-normalized reference signatures, signature refitting in individual tumors, and calculation of signature-attributed mutation burdens.

The accompanying manuscript applies SATS to 111,711 tumors from AACR Project GENIE to construct a real-world, panel-calibrated pan-cancer catalogue of targeted sequencing-derived mutational signatures. The package is intended for targeted-panel cohorts where whole-exome or whole-genome sequencing data are unavailable or not routinely generated.

Installation

The current source version in this repository is SATS v1.0.8. The recommended installation route is from the GitHub source tree:

```
if (!requireNamespace("devtools", quietly = TRUE))
  install.packages("devtools")
devtools::install_github("binzhulab/SATS", subdir = "source", upgrade = "never")
```

Alternatively, download SATS_1.0.8.tar.gz from the repository and install the source archive:

```
R CMD INSTALL SATS_1.0.8.tar.gz
```

The source archive can also be installed from within R:

```
install.packages("../SATS_1.0.8.tar.gz", repos = NULL, type = "source")
```

SATS was formerly available from CRAN. CRAN currently lists the package as archived, so the GitHub source installation above is the recommended installation route for the current version. After installation, load the package with:

```
library(SATS)
```

Required Inputs

SATS uses three main input matrices. The mutation catalogue matrix V has dimension $P \times N$, where rows are mutation contexts and columns are samples. For SBS analysis, $P = 96$. The panel-context matrix L has the same dimension as V and gives the number of mutation opportunities per million base pairs for each mutation context and sample. The reference signature matrix W has dimension $P \times K$, where columns are TMB-normalized reference signatures.

The row order of V , L and W must match. For SBS analyses, SATS supports the COSMIC-style 96-channel order and the `signer` order. The `SBS_order` argument in `GeneratePanelSize()` controls mutation-type ordering only; it does not select the COSMIC reference-signature version. The reference-signature version used by `MappingSignature()` is controlled separately by `COSMICv`, with "v3.4" as the current default.

SATS currently expects summarized mutation-count matrices, panel-context matrices and panel-coordinate data frames. It does not directly ingest raw VCF or BED files. VCF/BED-derived data should first be converted into mutation catalogue and panel-coordinate tables before running SATS.

Example Data

The package includes simulated data in the `SimData` object:

```
data(SimData, package = "SATS")
```

```
names(SimData)
```

`SimData$V` is a simulated 96 x 10027 SBS mutation catalogue matrix and `SimData$L` is the corresponding panel-context matrix. `SimData$TrueW_TMB` and `SimData$TrueH` are the simulated TMB-normalized signature profiles and activity matrix used to generate `SimData$V`. `SimData$PanelEx` and `SimData$PatientInfo` provide example panel and sample annotation data for constructing an L matrix.

Generating the Panel-Context Matrix

`GeneratePanelSize()` calculates panel-level mutation-context opportunity counts from panel-coordinate information. The input data frame must contain the columns `Chromosome`, `Start_Position`, `End_Position` and `SEQ_ASSAY_ID`. `Chromosome`, `Start_Position` and `End_Position` define the genomic interval, and `SEQ_ASSAY_ID` identifies the sequencing panel.

```
data(SimData, package = "SATS")

Panel_context <- GeneratePanelSize(
  genomic_information = SimData$PanelEx,
  Class = "SBS",
  SBS_order = "COSMIC",
  ref.genome = "hg19"
)
```

Class can be "SBS" or "DBS". For SBS analysis, `SBS_order` can be "COSMIC" or "signer". `ref.genome` can be "hg19" or "hg38" and requires the corresponding Bioconductor reference-genome package.

`GenerateLMatrix()` converts panel-level context counts into a sample-level L matrix by matching patient identifiers to sequencing panels. The `Patient_Info` data frame must contain `PATIENT_ID` and `SEQ_ASSAY_ID`.

```
L_mat <- GenerateLMatrix(Panel_context, SimData$PatientInfo)
dim(L_mat)
```

The resulting L matrix is used as the opportunity matrix for `signer()` and as the panel-context matrix for `EstimateSigActivity()` and `CalculateSignatureBurdens()`.

De Novo Signature Detection and Mapping

For cohort-level signature detection, SATS can be used with de novo profiles estimated by `signer` or another compatible signature extraction method. For large cohorts, samples may be grouped or pooled for computational feasibility before running de novo signature detection.

```
library(signer)

signer_re <- signer(M = V_sum, Opport = L_sum, nlim = c(1, 5))
W_hat <- signer_re$Phat
```

The de novo TMB-based profiles are then mapped to TMB-normalized reference signatures using `MappingSignature()`:

```
data(RefTMB, package = "SATS")

MappedSig <- MappingSignature(
  W_hat = W_hat,
  W_ref = RefTMB$TMB_SBS_v3.4
)
MappedSig
```

If `W_ref` is not supplied, `MappingSignature()` defaults to `COSMICv = "v3.4"` and uses `RefTMB$TMB_SBS_v3.4`. The returned data frame reports the selected reference signatures and the frequency with which each signature is selected across repeated penalized non-negative least-squares fits.

Estimating Signature Activities

After defining the reference signatures to use for refitting, estimate signature activities with `EstimateSigActivity()`.

```
data(SimData, package = "SATS")
data(RefTMB, package = "SATS")

SBS.list <- c("SBS1", "SBS2_13", "SBS4", "SBS5", "SBS6", "SBS89")
W_star <- as.matrix(RefTMB$TMB_SBS_v3.4[, SBS.list])

H_hat <- EstimateSigActivity(
  V = SimData$V,
  L = SimData$L,
  W = W_star
)
H_hat$H
```

`EstimateSigActivity()` uses an expectation-maximization algorithm and returns a list containing the estimated activity matrix H , the log-likelihood and a convergence flag. The estimated activity matrix has dimension $K \times N$.

Calculating Signature Burdens

Signature burdens are the expected numbers of mutations attributed to each selected signature in each tumor. They are calculated with `CalculateSignatureBurdens()`:

```
SigBdn <- CalculateSignatureBurdens(
  L = SimData$L,
  W = W_star,
  H = H_hat$H
)

round(SigBdn[, 1:5], 2)
```

The returned matrix has dimension $K \times N$, with signatures in rows and samples in columns.

Single-Tumor or Small-Cohort Refitting

SATS can also refit a prespecified signature set in a single tumor or a small cohort. In this setting, the reference set should be constrained to signatures that are relevant to the tumor type and detectable in targeted sequencing data. `RefTMB$SBS_refSigs` and `RefTMB$DBS_refSigs` provide cancer-specific SBS and DBS reference-signature lists.

```
data(SimData, package = "SATS")
data(RefTMB, package = "SATS")

SBS.list <- RefTMB$SBS_refSigs[
  RefTMB$SBS_refSigs$cancerType == "Skin Cancer or Melanoma",
  "COSMIC"
]
W_star <- as.matrix(RefTMB$TMB_SBS_v3.4[, SBS.list])

V1 <- SimData$SingleTumorEx[, "singleV", drop = FALSE]
L1 <- SimData$SingleTumorEx[, "singleL", drop = FALSE]

H_hat <- EstimateSigActivity(V = V1, L = L1, W = W_star)
SigBdn <- CalculateSignatureBurdens(L = L1, W = W_star, H = H_hat$H)
```

This workflow supports targeted-sequencing refitting when the set of signatures is known from a cancer-type-matched catalogue or from a prior cohort-level SATS analysis.

Tests, Workflow Example and Docker

Unit tests are provided in `source/tests/testthat/` for `CalculateSignatureBurdens()`, `EstimateSigActivity()` and `GeneratePanelSize()`. To run them locally:

```
cd source
Rscript -e 'testthat::test_dir("tests/testthat")'
```

The repository includes a minimal Nextflow example in `nextflow/example1/`. The example calls `GeneratePanelSize()` on an input `.rda` file containing `genomic_information` and optional `Class`, `SBS_order` and `ref.genome` objects. It is intended as a template for incorporating SATS panel-context generation into standardized workflow systems.

A Dockerfile is also provided for building an R environment with SATS and its core dependencies installed.

Citation and Web Resources

If you use SATS or the targeted-sequencing mutational-signature catalogue, please cite:

Lee et al., "A real-world pan-cancer catalogue of mutational signatures from 111,711 tumors" (submitted).

Interactive catalogue plots are available at <https://analysistools.cancer.gov/mutational-signatures/#!/catalog/STS>. An online signature refitting tool is available at <https://analysistools.cancer.gov/mutational-signatures/#!/refitting>.